

INTRODUCTION

More than 4,000 electric vehicles catch fire every year. These fires can take up to 24 hours to fully extinguish and are devastating. Clearly, electric vehicles can become very dangerous when they are poorly managed or poorly maintained. There are already methods in place to prevent these batteries from catching fire. In each EV, the **Battery Management System (BMS)** is a core piece of the prevention system; sensors combined with a central computer algorithm can control and monitor the battery cells.

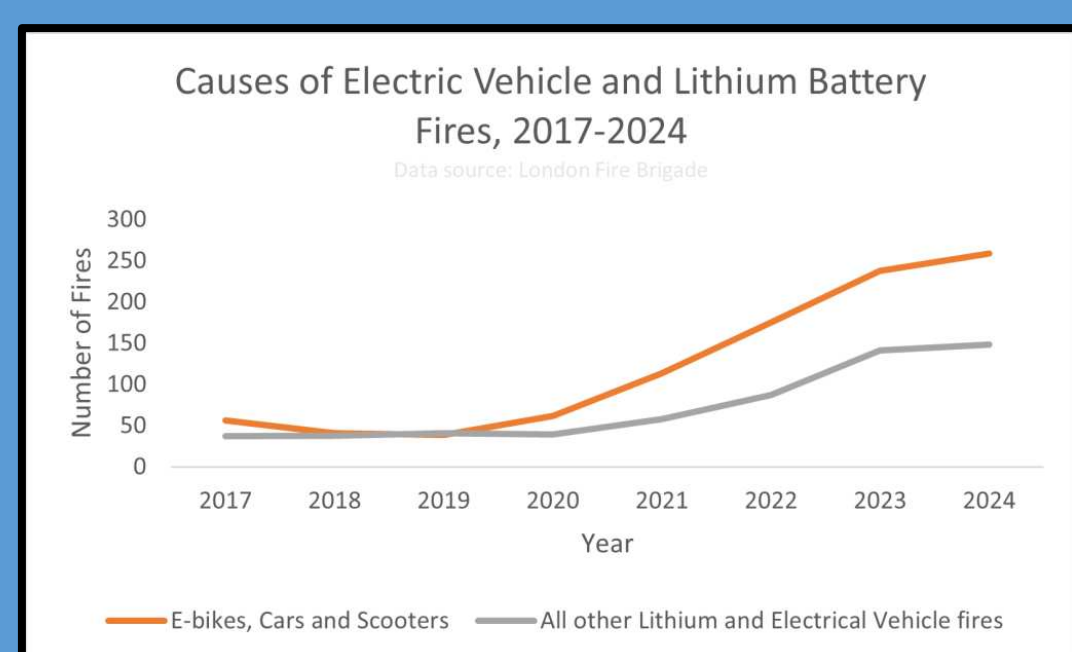


Figure 1: EV Battery fires over time in London.

The sensors read values like **voltage, current, temperature, etc.** and then send these values to a computer that evaluates and acts on these readings. Although these sensors can function in normal conditions, they are susceptible to degradation from elements such as **heat, cold, overvoltage, and overcurrent**. Because of this, they would then display inaccurate readings and could lead to poor battery management.

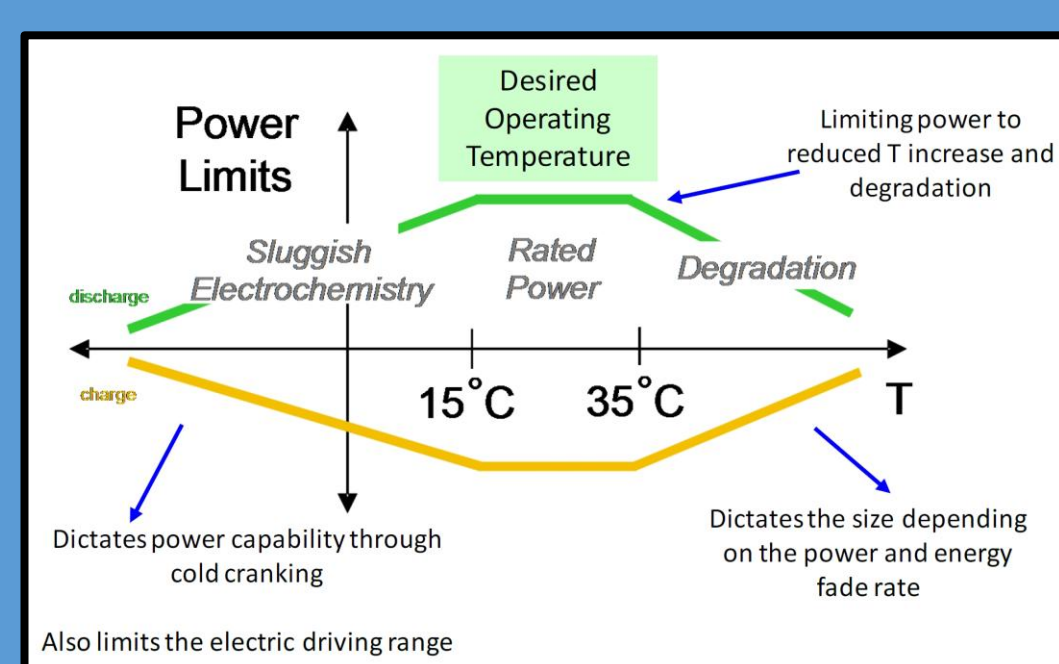


Figure 2: Power and Degradation in Temperature Range.

PURPOSE

Determining under what **conditions or stresses** and when do BMS sensors begin to malfunction is critical for developing preventive measures in the future. Creating a **desktop EV Battery cell emulator** is much safer than a physical battery, and it allows for test repetition and consistency. If the supply of **voltage, current, and the surrounding temperature** are highly varied, then primary reasons for fault will be identified from data collected.

Evaluating the Effect of Temperature, Voltage, and Current on EV Battery Sensor Performance

METHODS

1 Battery Cell Emulator Assembly

The Battery Emulator PCB and the other circuit board components were used to solder together an emulator circuit board. Then it was connected and configured through Arduino with the PC.

2 Cell Emulator Validation

Using a voltmeter and an ammeter, the voltage and current values coming out of the emulator were measured. These values were checked against the real-life values that were expected.

3 Sensor Integration and Assembly

Using the battery sensors, a breadboard system was made that would measure voltage, current, and temp., transmitting them to the PC. The integrated sensors measure the values of the emulator.

4 Sensor Control Group and Testing

The sensors were tested, giving a control group. The elements (voltage, current, temp.) were then varied in turn with the experimental group. The sensor data was read by the PC.

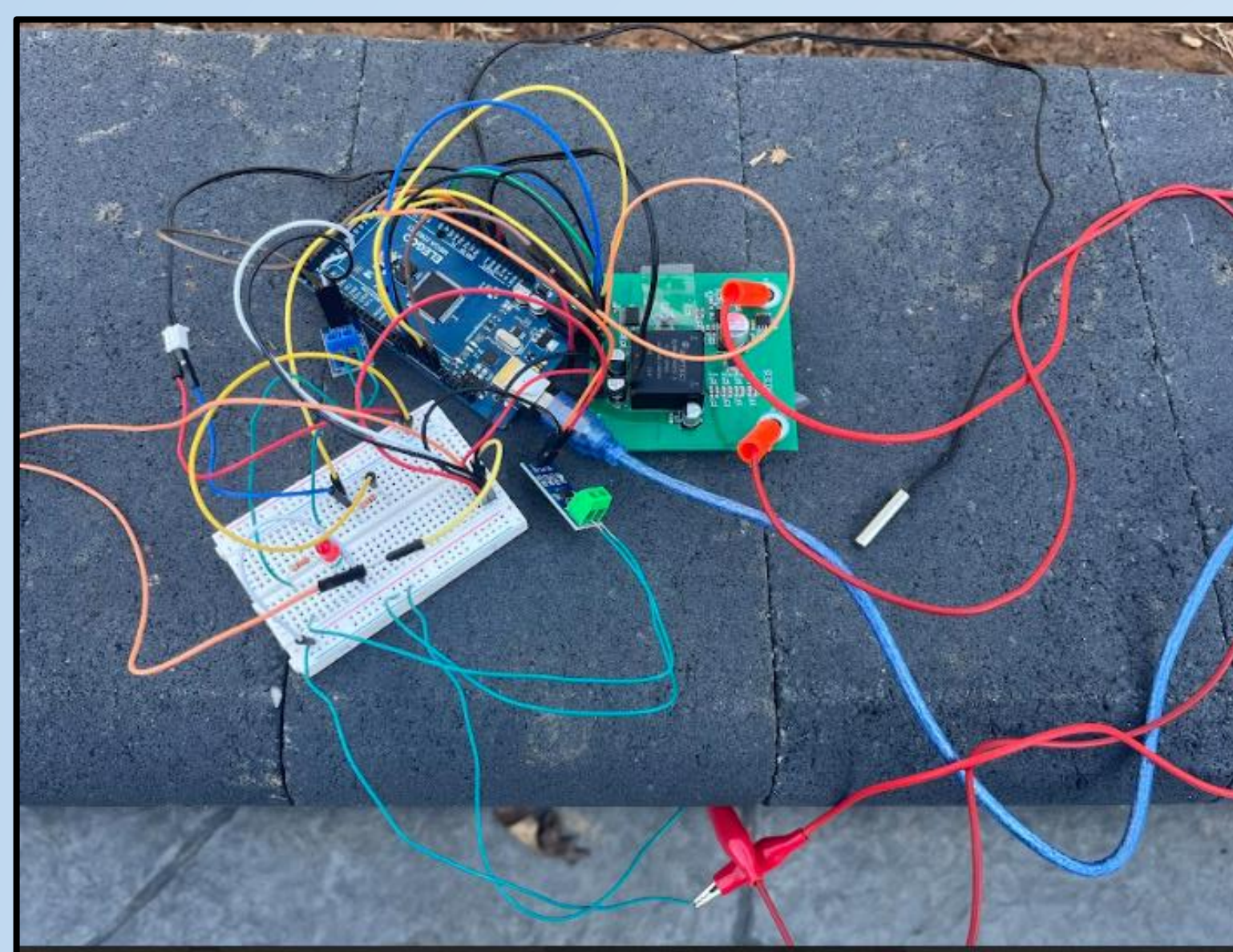


Figure 3: Battery Emulator and Sensor Array Final Prototype set-up.

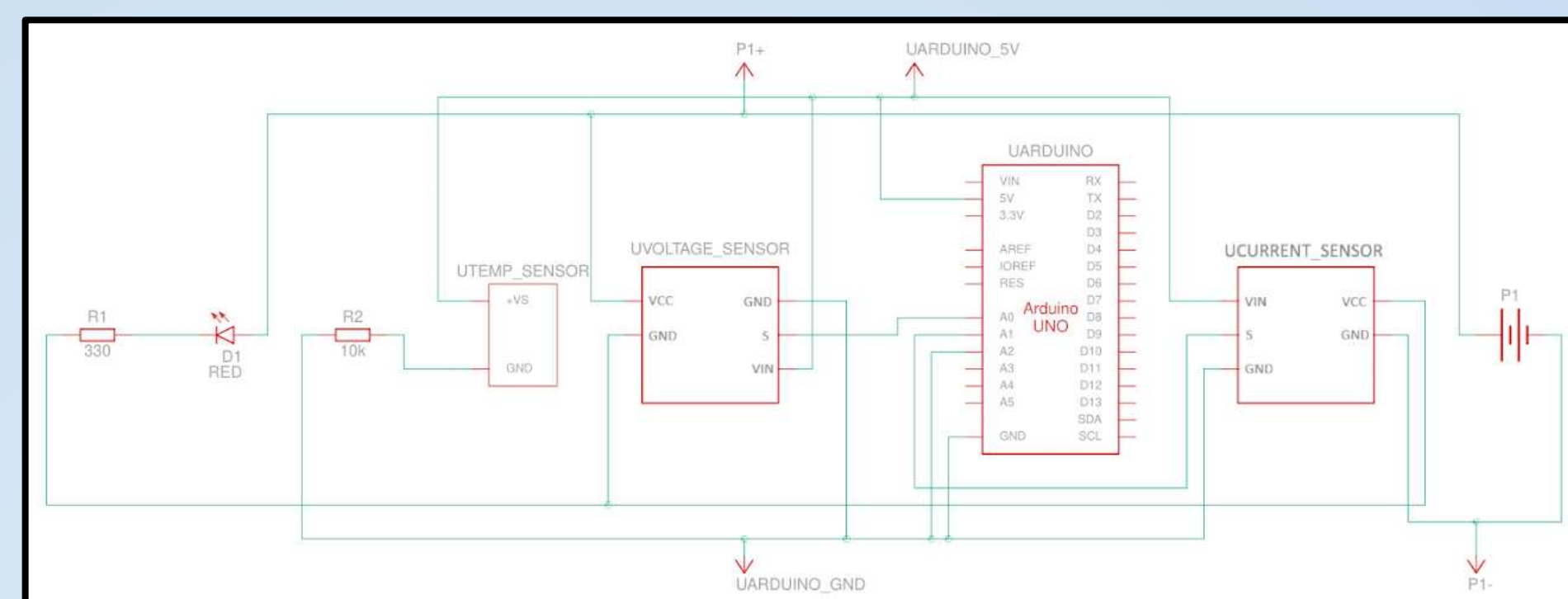


Figure 4: Final Prototype Electric Schematic.

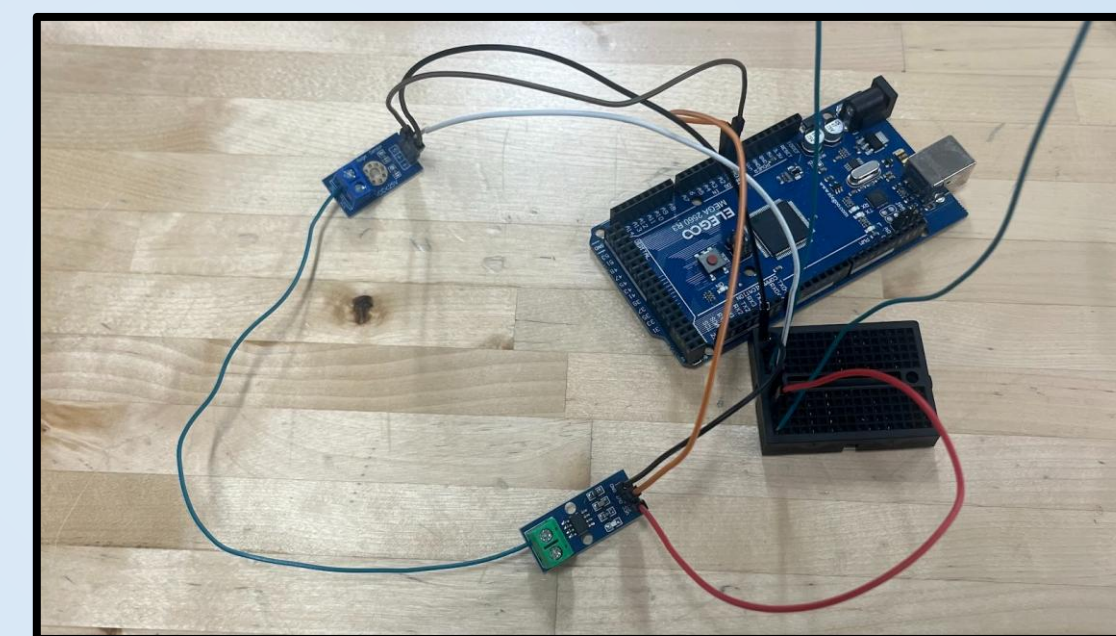


Figure 5: Prototyping Sensor System.

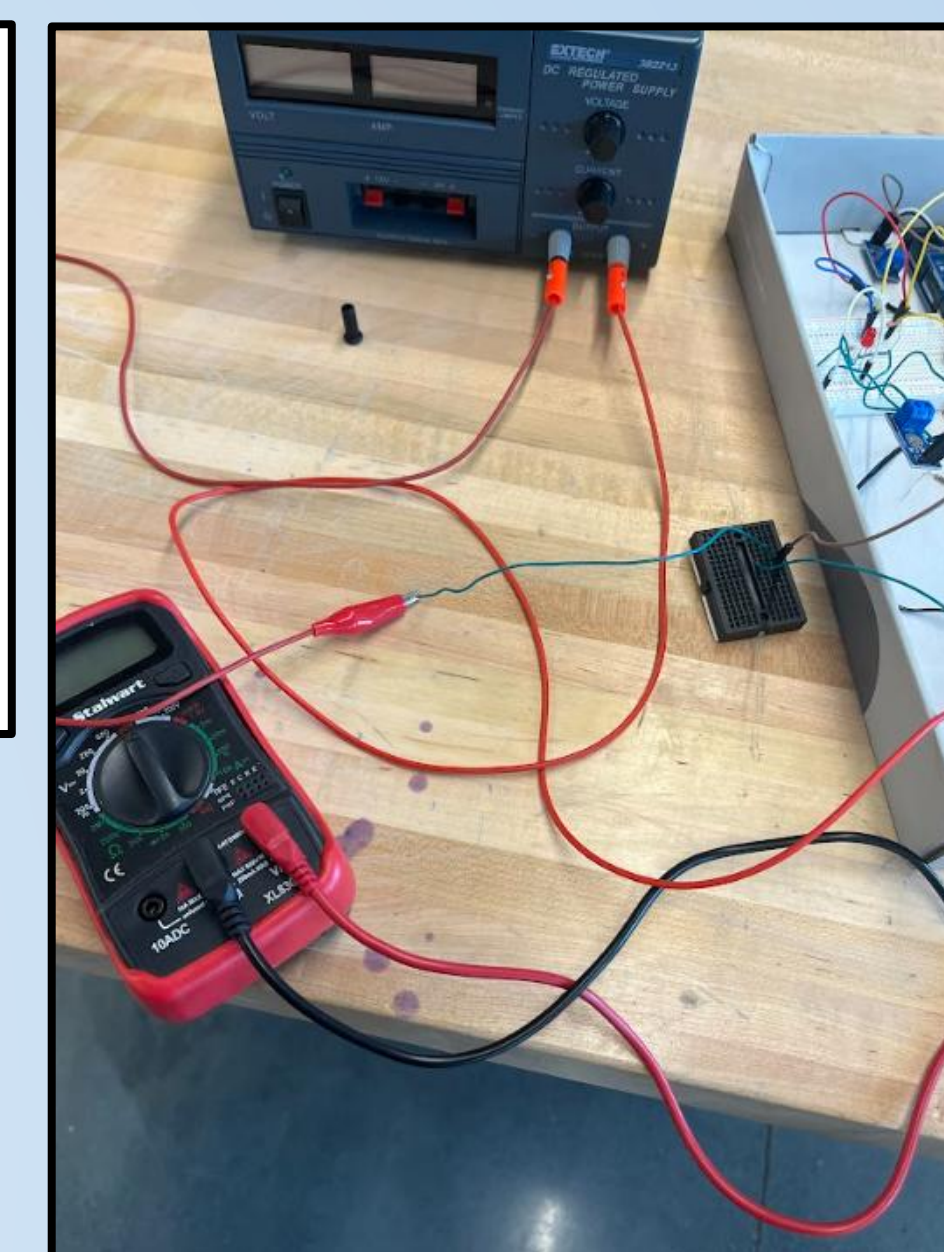


Figure 6: Final Prototype Voltage and Current Testing.

DATA COLLECTION AND RESULTS

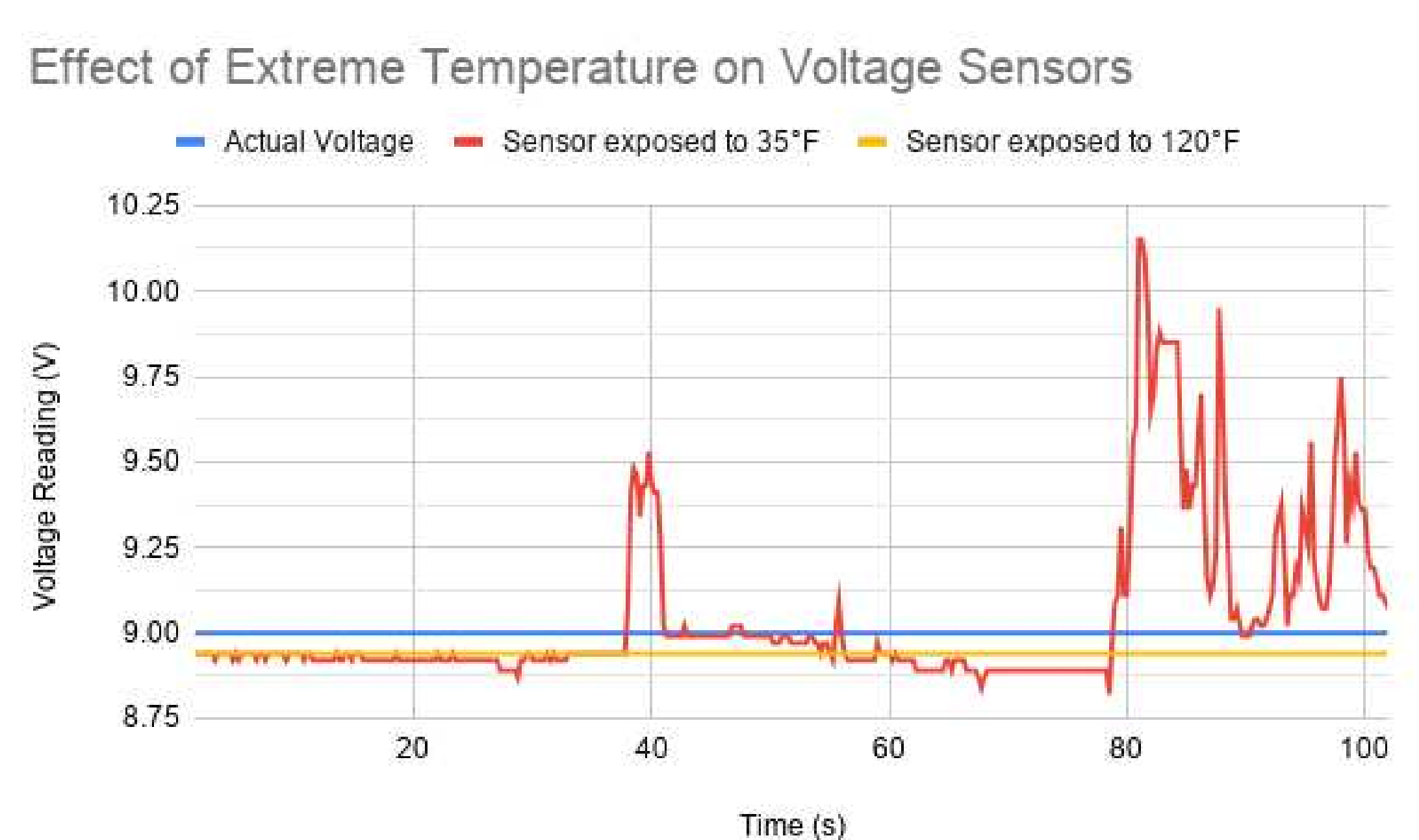


Figure 7: Graph of the Effect of Temperature on the Voltage Sensors.

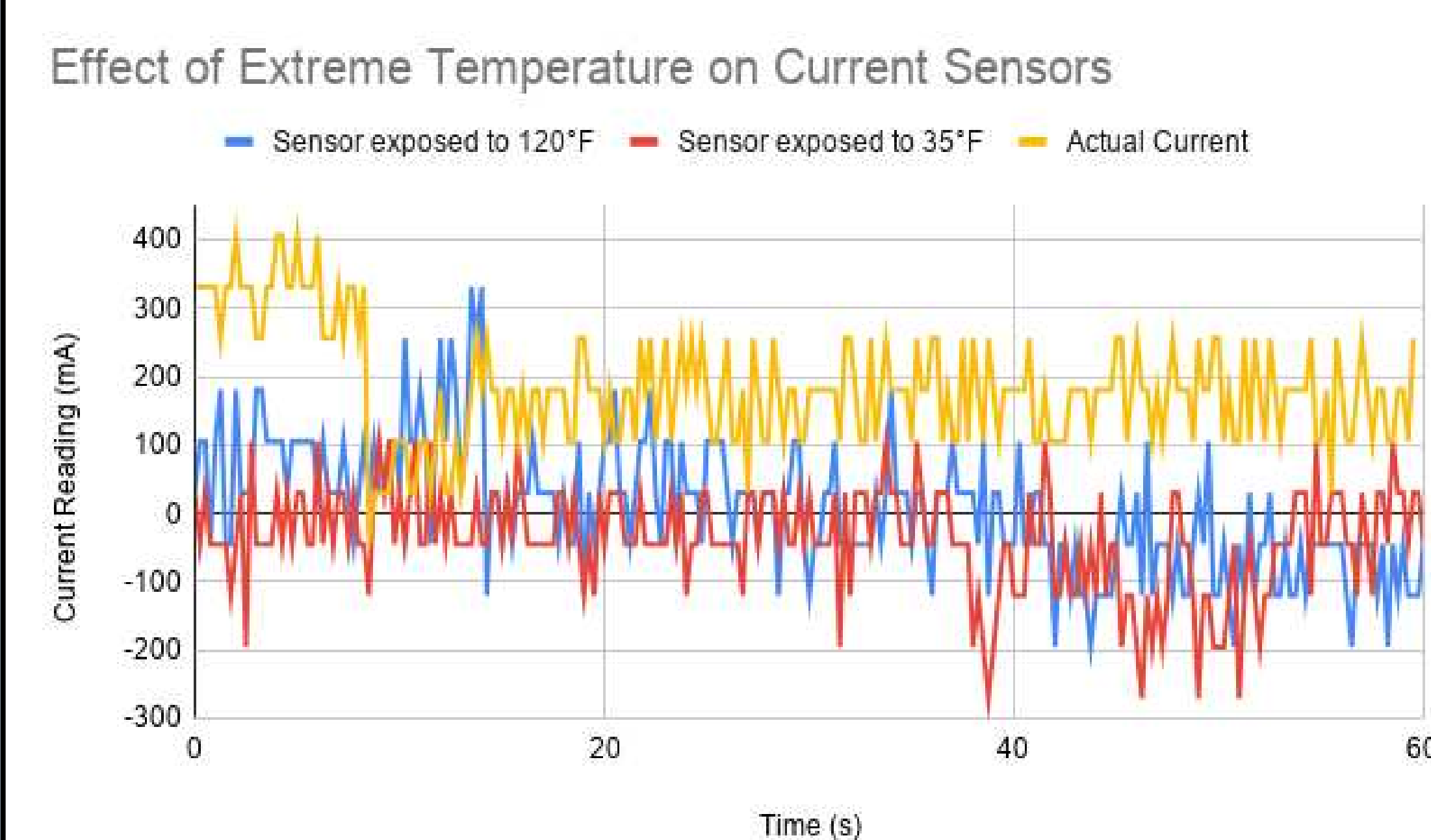


Figure 8: Graph of the Effect of Temperature on the Current Sensors.

Sensor:	Temp (°F)	Voltage (V)	Current (mA)
Avg. Reading Value for 5V supply (recommended, control)	70.15	8.94	184.17
Avg. Reading Value for 3V supply (below requirement)	46.18	8.90	-4511.51
Avg. Reading Value for 200Ω resistance (limiting current supply)	49.23	8.91	-10193.85
Readings after returning to recommended supply	70.1	8.94	180.7

CONCLUSION

According to the data, extreme conditions or environments have a **significant effect** on most of the sensor readings. However, these malfunctions were all **proven to be temporary**; after the sensors returned to a nominal environment, their readings had **returned to accuracy**.

With respect to temperature, the current and voltage sensors were most affected by very cold temperatures, displaying values that are at times over **12% erroneous for voltage** and over **78% for current**. With respect to low supply of current and voltage, the current sensor and the temperature sensor exhibited the worst malfunctions, with readings about **39% less** for temperature and over **185% difference** for current. These data points were the most significant ones; other conditions did not impact readings as much.

APPLICATIONS AND EXTENSIONS

With this information, this project can be taken in a new direction: **the prevention of these malfunctions**. Knowing exactly how the sensors fail can allow future research to explore measures to keep the sensors away from these proven adverse elements. For example, research could aim to **insulate** sensors against cold temperatures or place **safe-guards in the circuit** to prevent an excess supply of voltage.



Figure 5: A Fuse that could be used to protect the sensor in a future circuit.

REFERENCES

- Di Rienzo, R., Verani, A., Baronti, F., Roncella, R., & Saletti, R. (2022). Modular Battery Emulator for Development and Functional Testing of Battery Management Systems: The Cell Emulator. *Electronics*, 11(8), 1215. <https://doi.org/10.3390/electronics11081215>
- Rezvanizadeh, S. M., Liu, Z., Chen, Y., & Lee, J. (2014). Review and recent advances in battery health monitoring and prognostics technologies for electric vehicle (EV) safety and mobility. *Journal of Power Sources*, 256, 110–124. <https://doi.org/10.1016/j.jpowsour.2014.01.085>
- Zhao, Y., Geng, L., Shan, S., Du, Z., Hu, X., & Wei, X. (2024). Review of sensor fault diagnosis and fault-tolerant control techniques of lithium-ion batteries for electric vehicles. *Journal of Traffic and Transportation Engineering (English Edition)*. <https://doi.org/10.1016/j.jtte.2024.09.003>